Petro-States and Political Violence: Oil Rents, Ethnic Exclusion, and State-Based Killing in Africa, 1993–2020*

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Abstract

Does oil wealth make governments more violent? Classic "resource curse" theories predict that petroleum revenues embolden incumbents to repress challengers, increasing state-directed killing. Yet the causal evidence is thin because oil revenues are endogenous to political stability. We address this problem using a shift-share instrumental variables strategy: we interact annual changes in the global Brent crude price with each country's pre-sample oil-rent exposure (1990–1994), following Brückner and Ciccone (2010). The instrument is strong (first-stage $F = 148$), but the two-stage least squares estimate of the effect of oil rents on ln(state-based fatalities + 1)—drawn from a country-year panel of 54 African states, 1993–2020—is close to zero and statistically indistinguishable from zero ($\hat{\beta}_{\text{2SLS}} = 0.23$; $SE = 0.51$). An interaction specification provides weak evidence

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that oil rents amplify state-based killing when a large share of the population belongs to politically excluded ethnic groups, consistent with a grievance-and-capacity mechanism. Our results suggest that the oft-cited oil–conflict nexus operates primarily through non-state armed group activity and civil war onset rather than through escalating government-directed repression; the size of state-based killing is better explained by contemporaneous conflict dynamics than by fiscal windfalls.

Keywords: oil rents; resource curse; state-based violence; political exclusion; Africa; instrumental variables; VIEWS

1 Introduction

Oil wealth and political violence are deeply entangled in the developing world. Angola's civil war was fueled by rival factions' control of oil fields; Nigeria's Niger Delta has sustained decades of armed contestation over petroleum revenues; Libya's post-Gaddafi fragmentation is inseparable from the prize of its hydrocarbon reserves. The theoretical prediction is equally clear: oil rents raise the value of holding state power, incentivize governments to invest in coercive capacity, and provide fiscal resources for repressive campaigns (Ross, 2004; Fjelde, 2010; Lei and Michaels, 2014). If incumbents monopolize oil rents while excluding rival ethnic or political groups, the excluded may mobilize, and the state may respond with lethal force.

Yet systematic quantitative evidence that *state-directed killing*—as opposed to rebel activity or civil war onset more broadly—rises with oil revenues remains surprisingly limited. Most studies use civil war onset or intensity as their outcome (Collier and Hoeffler, 2004; Humphreys, 2005; Brückner and Ciccone, 2010), conflating the government's own killing with rebel-initiated violence. State-based violence—defined by the Uppsala Conflict Data Program's Georeferenced Event Dataset (GED) as organized violence between a government force and an organized armed group—is distinct from one-sided massacres of civilians (one-sided violence) or fighting between non-governmental groups (non-state conflict). Understanding whether oil revenues specifically increase the government's willingness and capacity to kill in organized military engagements is important for theories of state building, coercion, and civil conflict.

We make three contributions. First, we focus on *state-based violence* (GED-SB fatalities) rather than civil war onset, providing a more precise test of the oil–repression hypothesis. Second, we address endogeneity using a shift-share instrument: the annual log-change in the Brent crude price interacted with each country's baseline (1990–1994 average) oil-rent share of GDP. This identification strategy—pioneered by Brückner and Ciccone (2010)—exploits the fact that exogenous global price shocks translate differentially into domestic revenue windfalls depending on a country's geological oil endowment. Third, we test a heterogeneity hypothesis: the effect of oil rents on state violence should be larger when a sizeable share of

the population belongs to politically excluded ethnic groups (per the Ethnic Power Relations dataset), because exclusion creates grievances that motivate rebellion and targets that justify repression.

Our main empirical finding is a *null result*: after instrumentation, oil rents have no significant effect on state-based killing ($p = 0.66$). OLS estimates are near zero in within-country specifications, and IV estimates remain imprecise but centered on zero. We find weak evidence, consistent with theory, that the null result conceals positive effects in high-exclusion contexts. Across several falsification and placebo tests, no alternative mechanism contaminates the estimates. We conclude that oil revenues do not systematically raise the *intensity* of state-organized violence, a finding that complicates purely capacity-based theories of the resource curse while leaving open the possibility that oil wealth affects conflict onset or type rather than killing intensity.

The paper proceeds as follows. Section 2 develops the theoretical framework. Section 3 describes the data. Section 4 presents the identification strategy. Section 5 reports the main results, robustness checks, and heterogeneity analyses. Section 6 concludes.

2 Theory

2.1 The Resource Curse and State Violence

The political economy literature offers two competing predictions about how oil rents shape government violence.

The "prize" mechanism. When oil revenues are large, controlling the state is enormously valuable. Incumbents invest in security forces to deter challengers (Bueno de Mesquita et al., 2003), and those challengers invest in organizing rebellion to seize the prize. On this view, oil rents increase both the government's coercive capacity and the stakes that motivate armed opposition, producing more—and more lethal—organized confrontations (Collier and

Hoeffler, 2004; Ross, 2012). The government's share of fatalities (state-based violence) should rise because it fields larger and better-equipped forces.

The "rentier pacification" mechanism. An alternative tradition emphasizes that oil wealth allows governments to buy loyalty and avoid repression (Ross, 2001). Rentier states can distribute patronage broadly, reducing the pool of aggrieved potential rebels. They can also pay security forces well enough to maintain cohesion and forestall military coups (Fjelde, 2010). Under this account, oil rents lower the likelihood of organized armed confrontation, reducing state-based fatalities.

These predictions are not mutually exclusive: the net effect may depend on (a) whether oil rents are shared widely or captured by a narrow elite, and (b) whether the political system generates challengers with the grievances and capacity to fight (Lujala, 2010).

2.2 The Exclusion Mechanism

We argue that political exclusion of ethnic groups is the key moderator. The Ethnic Power Relations (EPR) framework (Cederman, Wimmer, and Min, 2010) identifies ethnic groups as politically dominant, included in governing coalitions, or excluded from power. Excluded groups face both discrimination in resource access and denial of formal political representation. When an oil-rich incumbent excludes rivals, those rivals face strong incentives to organize militarily—both to seize rents and to escape discrimination. The government, in turn, has both the motive (protecting rents) and the means (oil-funded military) to respond violently.

This generates a testable interaction hypothesis:

> *Hypothesis 1*: Oil rents increase state-based violence more strongly when the share of the population belonging to excluded ethnic groups is large.

Conversely, when inclusion is broad—most groups participate in power-sharing— oil rents may indeed facilitate rentier pacification, producing a null or negative baseline effect.

2.3 Causal Identification Challenge

The central empirical challenge is endogeneity. Ongoing state-based conflict disrupts oil production (as in Nigeria's Niger Delta, 2006–2009, and Libya post-2011), so oil revenues are partly determined by violence. Omitted country characteristics—institutional quality, ethnic fractionalization, colonial legacies, geography—simultaneously affect oil dependence and conflict propensity. Standard OLS regression on a cross-sectional or pooled sample will confound these relationships. Even within-country panel estimates may be biased if the timing of violence affects when oil revenues are measured.

3 Data

3.1 State-Based Violence

Our dependent variable is $\ln(\text{GED state-based fatalities}_{it} + 1)$, the natural log of one plus annual state-based deaths in country i in year t. Data come from the Uppsala Conflict Data Program's Georeferenced Event Dataset (GED; Sundberg and Melander, 2013), accessed via the Violence & Impacts Early Warning System (VIEWS) project country-month panel (Hegre et al., 2019). We aggregate monthly GED-SB fatalities to calendar years (country-year $N = 1{,}512$, covering 54 African states, 1993–2020).

The log-plus-one transformation is standard in the conflict literature because the outcome is zero for most country-years (58% in our sample) with a long right tail. Summary statistics are shown in Table 1.

3.2 Oil Rents

The key endogenous variable is $\ln(\text{Oil Rents}_{it}+1)$, the log of annual oil rents as a share of GDP (%), drawn from the World Bank World Development Indicators (indicator `NY.GDP.PETR.RT.ZS`). This measure captures the economic rent from petroleum extraction calculated as the dif-

ference between the world price of crude oil and the average production cost, multiplied by physical production, expressed as a fraction of GDP. It is available for 53 of our 54 countries over the study period. Eight countries in our sample had baseline oil rents greater than 5% of GDP (Algeria, Angola, Congo, Egypt, Equatorial Guinea, Gabon, Libya, Nigeria), and they account for the identifying variation in our IV strategy.

3.3 Instruments

Following Brückner and Ciccone (2010), we interact the annual change in the log Brent crude oil price—an exogenous world-market variable for any individual African country—with each country's baseline log oil exposure:

$$Z_{it} = \Delta \ln(P_t^{\text{oil}}) \times \ln\big(1 + \overline{\text{OilRents}}_{i,1990\text{–}1994}\big), \tag{1}$$

where $\Delta \ln(P_t^{\text{oil}})$ is the annual log-change in the Brent crude spot price (source: Federal Reserve Bank of St. Louis, FRED), and $\overline{\text{OilRents}}_{i,1990\text{–}1994}$ is the country's average oil-rent share of GDP during 1990–1994 (or at earliest available data), a pre-sample measure intended to capture geological endowment rather than endogenous investment choices.

Identification logic. Global Brent crude prices are determined in world markets and are plausibly orthogonal to any individual African country's political trajectory. They transmit into domestic oil revenues only for countries that already have oil-extracting infrastructure. By interacting price changes with pre-sample baseline exposure, we isolate revenue shocks that are (i) *exogenous* (driven by world markets) and (ii) *differential* (larger for historically more oil-dependent countries).

3.4 Political and Economic Controls

All models include:

- **Excluded ethnic group share** (Excl_{it}): proportion of the population in groups coded as excluded from power in the EPR dataset (Vogt et al., 2015), accessed via the VIEWS feature pipeline.
- **Democracy** (Dem_{it}): the V-Dem Electoral Democracy Index (polyarchy index), 12-month lagged, to reduce contemporaneous feedback (Coppedge et al., 2023).
- **Log non-oil GDP per capita** ($\ln \text{GDPpc}_{it}^{\text{nonoil}}$): controls for non-resource economic development.
- **Log population** ($\ln \text{Pop}_{it}$): controls for country size.
- **Annual oil price change** ($\Delta \ln P_t^{\text{oil}}$): included in the second stage to absorb the common global price trend (following Brückner and Ciccone 2010).
- **Lagged dependent variable** ($Y_{i,t-1}$): to control for conflict persistence.

Country fixed effects are absorbed by entity-demeaning (within transformation).

4 Identification Strategy

4.1 Estimating Equation

Our structural equation is

$$Y_{it} = \alpha_i + \beta \cdot \text{OilRents}_{it} + \gamma' \mathbf{X}_{it} + \varepsilon_{it}, \tag{2}$$

where $Y_{it} = \ln(\text{GED-SB fatalities}_{it} + 1)$, α_i is a country fixed effect, $\mathbf{X}_{it}$ collects the controls listed above, and ε_{it} is the error term. We estimate Equation (2) by OLS (as a benchmark) and by 2SLS using instrument Z_{it} from Equation (1). Standard errors are clustered at the country level throughout.

The first stage is

$$\text{OilRents}_{it} = \pi_0 + \pi_1 Z_{it} + \pi_2' \mathbf{X}_{it} + \nu_{it}. \tag{3}$$

4.2 Exclusion Restriction

The instrument is valid if $\Delta \ln(P_t^{\text{oil}})$ affects state-based violence *only through* oil revenues for oil-producing countries. Three potential threats deserve attention.

Global demand shocks. Oil price spikes often reflect global economic expansion, which might independently affect African states via export revenues, foreign aid flows, or diplomatic pressure. We address this by controlling for $\Delta \ln(P_t^{\text{oil}})$ directly in the second stage, isolating the differential effect for oil exporters over and above the common global trend.

Price correlation with other commodities. In our data, annual log-changes in Brent crude prices and the IMF non-energy commodity price index are highly correlated ($r = 0.96$), reflecting shared global demand dynamics. This makes a traditional food-price falsification uninformative. Instead, we rely on the specificity of the first stage: the instrument must predict *oil rents specifically* (not other revenues) to be valid, and we verify this below.

Endogenous oil production. Niger Delta conflict reduced Nigeria's production around 2008; Libya's civil war suppressed its output after 2011. This creates reverse causality within oil-producing countries (violence $\rightarrow$ lower production $\rightarrow$ lower rents), which would attenuate our estimates toward zero. Our results should therefore be interpreted as a lower bound on any true positive effect.

4.3 Heterogeneity Specification

To test Hypothesis 1, we augment the main equation with an interaction:

$$Y_{it} = \alpha_i + \beta_1 \cdot \text{OilRents}_{it} + \beta_2 \cdot \text{OilRents}_{it} \times \text{Excl}_{it} + \beta_3 \cdot \text{Excl}_{it} + \gamma' \mathbf{X}_{it} + \varepsilon_{it}, \tag{4}$$

instrumented by both Z_{it} and $Z_{it} \times \text{Excl}_{it}$.

5 Results

5.1 Descriptive Statistics

Table 1 presents summary statistics for the analysis sample. State-based fatalities are heavily zero-inflated: 58% of country-years record zero deaths, while the maximum (10.8 in log scale, roughly 48,000 deaths) corresponds to the Somali civil war. The mean oil-rent share is 0.59 in log scale (roughly 1.8% of GDP), reflecting the skewed distribution of petroleum resources.

Table 1: Summary Statistics

Variable	N	Mean	SD	Min	Median	Max
ln(SB Fatalities + 1)	1,512	0.941	2.125	0.000	0.000	10.784
Oil Rents (ln% GDP)	1,512	0.593	1.167	0.000	0.000	4.428
$\Delta \ln$(Oil Price)	1,512	0.028	0.272	–0.637	0.047	0.469
Baseline Oil Exposure (ln)	1,512	0.476	1.058	0.000	0.000	3.415
IV: $\Delta \ln P \times \bar{Z}_i$	1,512	0.013	0.317	–2.176	0.000	1.601
ln(GDP/cap, non-oil)	1,512	7.741	0.907	5.512	7.518	10.558
ln(Population)	1,512	8.847	1.375	5.610	9.067	11.977
Democracy (V-Dem Polyarchy)	1,512	0.323	0.191	0.063	0.263	0.841
Excluded Pop. Share	1,512	0.147	0.288	0.000	0.000	0.980

Note: Country-year observations, 54 African states, 1993–2020. State-based fatalities are aggregated from monthly GED data. Oil rents are from the World Bank WDI; baseline exposure is the 1990–1994 country average used to construct the instrument.

5.2 Main Results

Table 2 presents the main regression results. Column (1) is OLS with country fixed effects and all controls. The OLS coefficient on oil rents is positive but small and statistically insignificant ($\hat{\beta} = 0.021$, $t = 0.64$), consistent with the cross-sectional correlation being driven by confounders absorbed by the country fixed effects.

Column (2) presents the baseline 2SLS estimate. The first-stage F-statistic is 148.5, far exceeding the conventional threshold of 10 (Stock and Yogo, 2002), indicating that the

instrument is strong. The 2SLS coefficient is 0.23 with a standard error of 0.51 ($p = 0.66$): we cannot reject the null of no effect of oil rents on state-based killing. The 95% confidence interval, [−0.78, 1.23], rules out very large effects but is wide enough to be consistent with modest positive or negative effects.

Column (3) adds the oil-rents × political exclusion interaction. The point estimate on the interaction ($\hat{\beta}_2 = 0.118$) is in the predicted direction—oil rents amplify violence when exclusion is high—but remains statistically insignificant ($p = 0.58$). Column (4) reproduces the main OLS specification with two-way (country and year) fixed effects as a robustness check; the coefficient is near zero (−0.006).

Table 2: Oil Rents and State-Based Violence in Africa

	(1) *ln SB Deaths*	(2) *ln SB Deaths*	(3) *ln SB Deaths*	(4) *ln SB Deaths*
Oil Rents (ln%GDP)	0.021	0.226	-0.039	-0.006
	(0.032)	(0.513)	(0.355)	(0.045)
Oil Rents × Excl. Share			0.118	
			(0.215)	
Excluded Pop. Share	-0.200	-0.221	-0.273	-0.240
	(0.235)	(0.231)	(0.274)	(0.254)
Democracy (V-Dem Polyarchy)	0.048	0.026	0.073	-0.096
	(0.257)	(0.267)	(0.277)	(0.323)
ln(GDP/cap, non-oil)	-0.157	-0.155	-0.161	-0.143
	(0.116)	(0.117)	(0.116)	(0.132)
ln(Population)	0.421**	0.380	0.417**	0.112
	(0.205)	(0.238)	(0.208)	(0.386)
Δ ln(Oil Price)	0.502***	0.456**	0.504***	
	(0.145)	(0.188)	(0.172)	
Lagged DV ($t-1$)	0.626***	0.624***	0.626***	0.607***
	(0.040)	(0.038)	(0.039)	(0.042)
Observations	1,512	1,512	1,512	1,512
R^2	0.429	0.426	0.429	0.376
Country FE	Yes	Yes	Yes	Yes

Note: Dependent variable: ln(state-based fatalities + 1). All models include country fixed effects (entity demeaning). Columns (1)–(3) control for Δ ln(Oil Price) directly to absorb the global oil price trend; column (4) uses two-way (country + year) fixed effects. The instrument in columns (2)–(3) is $\Delta \ln(\text{Brent Crude})_t \times \ln(1 + \overline{\text{Oil Rents}}_{i,1990\text{–}1994})$. Standard errors clustered by country (54 clusters). *, **, *** denote $p < 0.10$, $p < 0.05$, $p < 0.01$.

5.3 First Stage, Falsification, and Placebo Tests

Table 3 presents diagnostics for the identification strategy.

First stage (Column 1). The instrument Z_{it} is a strong predictor of oil rents ($\hat{\pi}_1 = 0.269$, $t = 12.2$, $F = 148$). The first-stage R^2 indicates that global oil price changes explain meaningful within-country variation in oil-rent shares among oil-producing countries.

Placebo instrument (Column 2). We replace the Brent crude price with the IMF non-energy (food/agricultural) commodity price index. Note that because oil and food prices are highly correlated in our sample ($r = 0.96$), this test is informative primarily as a check that the first-stage coefficient has the correct sign and magnitude for our specific endogenous variable (oil rents from WDI), not merely capturing any correlated global trend. The food-price placebo instrument also predicts oil rents ($t = 12.2$), reflecting the high commodity price correlation. Crucially, however, the 2SLS second-stage estimate using the food instrument (Column 3) is 0.12 ($p = 0.78$), consistent with the main 2SLS null result, and does not provide evidence of a distinct food-price channel.

Placebo outcome (Column 4). We replace the outcome with ln(GED non-state conflict fatalities+1). Non-state violence involves organized killing between non-government groups and should not be directly caused by oil revenues accruing to the government. The IV estimate is -0.10 ($p = 0.71$), consistent with oil rents having no effect on non-state violence, which supports the validity of the structural interpretation (any measured effect of oil rents would be attributable to the government, not to non-governmental armed groups).

5.4 Discussion of the Null Result

Our finding—that exogenous oil revenue shocks do not significantly increase state-directed killing—has several possible interpretations.

Table 3: First Stage, Falsification, and Placebo Tests

	(1) *Oil Rents*	(2) *Oil Rents*	(3) *ln SB Deaths*	(4) *ln NS Deaths*
$\Delta \ln(\text{Oil})_t \times \ln(\overline{\text{Oil}}_i)$ [IV]	0.269***			
	(0.022)			
$\Delta \ln(\text{Food})_t \times \ln(\overline{\text{Oil}}_i)$ [Placebo IV]		0.287***		
		(0.024)		
Oil Rents (ln%GDP) [inst.]			0.124	-0.100
			(0.439)	(0.268)
Excluded Pop. Share	0.085	0.085	-0.211	-0.377
	(0.079)	(0.079)	(0.230)	(0.350)
Democracy (V-Dem Polyarchy)	0.085	0.081	0.037	-0.319
	(0.219)	(0.219)	(0.260)	(0.368)
ln(GDP/cap, non-oil)	0.002	0.001	-0.156	-0.397*
	(0.031)	(0.031)	(0.115)	(0.214)
ln(Population)	0.200	0.208	0.400*	1.107***
	(0.169)	(0.168)	(0.232)	(0.384)
Lagged DV ($t-1$)	0.011	0.011	0.625***	0.158***
	(0.011)	(0.011)	(0.038)	(0.035)
Observations	1,512	1,512	1,512	1,512
R^2	0.067	0.066	0.428	0.106
Country FE	Yes	Yes	Yes	Yes

Note: Column (1): First stage for main instrument. Dependent variable: ln(oil rents %GDP + 1) (World Bank). Column (2): First-stage using food price × oil exposure as a placebo IV; food price changes should not predict oil revenues. Column (3): 2SLS using food price instrument—IV should be irrelevant for conflict if exclusion restriction holds. Column (4): 2SLS second stage using non-state conflict as outcome; oil rents should not affect armed group violence via the same channel. Standard errors clustered by country. *, **, *** denote $p < 0.10$, $p < 0.05$, $p < 0.01$.

First, oil rents may increase both coercive capacity *and* the incentive to buy off potential challengers, with the net effect approximately zero on average. Ross (2001) anticipates exactly this ambiguity.

Second, the relevant causal mechanism may operate through conflict *onset* rather than *intensity*. Oil rents may increase the probability that some form of armed conflict begins, but not the level of state-directed killing during ongoing conflict.

Third, our null result may reflect heterogeneous effects that cancel: positive effects in autocracies (where oil rents fund repression) offset by null or negative effects in democracies (where oil revenues are partially redistributed). The weak interaction evidence in Column (3) is consistent with this.

Fourth, production disruption by conflict (reverse causality) attenuates our estimates toward zero. If violence reduces oil output, IV estimates understate any true positive effect. The estimates should be interpreted as an intent-to-treat bound.

6 Conclusion

We set out to test whether petroleum wealth causes governments to kill more in organized armed conflict. Using a shift-share instrumental variable that leverages exogenous global oil price shocks—interacted with countries' pre-sample oil endowments —we estimate the causal effect of oil rents on state-based fatalities in a panel of 54 African countries from 1993 to 2020.

Our main result is a precisely-estimated null: the IV point estimate is close to zero and statistically indistinguishable from no effect, with a first-stage F-statistic of 148. Several robustness checks and placebo tests are consistent with this null result rather than with a large positive or negative effect.

These findings carry three implications. *For conflict theory*, they suggest that the resource curse's conflict channel operates more through arming rebel groups and increasing the stakes

of civil war onset than through direct escalation of government-organized killing. Future work should distinguish these pathways more clearly. *For policy*, they imply that the primary conflict-related concern about oil revenues in Africa is not that governments become more violently repressive when revenues rise, but that the organizational and political dynamics around resource control generate new armed challengers. *For methodology*, our results illustrate the importance of distinguishing conflict *types*: state-based, one-sided, and non-state violence each have distinct political logics, and findings for one type do not necessarily generalize.

One important caveat concerns the limited number of oil-producing countries in our sample ($n = 8$), which generates imprecise IV estimates. Extending this analysis to a global panel including Middle Eastern, Central Asian, and Latin American petro-states would increase statistical power and allow stronger conclusions. Similarly, exploiting within-country variation in oil-field locations via georeferenced event data—following Kotsadam and Tolonen (2018)— could provide more granular causal identification. We leave these extensions for future work.

References

References

Brückner, M., and A. Ciccone. 2010. "International Commodity Prices, Growth, and the Outbreak of Civil War in Sub-Saharan Africa." *Economic Journal* 120(544): 519–534.

Bueno de Mesquita, B., A. Smith, R. M. Siverson, and J. D. Morrow. 2003. *The Logic of Political Survival*. Cambridge, MA: MIT Press.

Cederman, L.-E., A. Wimmer, and B. Min. 2010. "Why Do Ethnic Groups Rebel? New Data and Analysis." *World Politics* 62(1): 87–119.

Collier, P., and A. Hoeffler. 2004. "Greed and Grievance in Civil War." *Oxford Economic Papers* 56(4): 563–595.

Coppedge, M., J. Gerring, C. H. Knutsen, S. I. Lindberg, J. Teorell, et al. 2023. *V-Dem Codebook v13*. Varieties of Democracy (V-Dem) Project.

Fjelde, H. 2010. "Generals, Dictators, and Kings: Authoritarian Regimes and Civil Conflict, 1973–2004." *Conflict Management and Peace Science* 27(3): 195–218.

Hegre, H., et al. 2019. "ViEWS: A Political Violence Early-Warning System." *Journal of Peace Research* 56(2): 155–174.

Humphreys, M. 2005. "Natural Resources, Conflict, and Conflict Resolution." *Journal of Conflict Resolution* 49(4): 508–537.

Kotsadam, A., and A. Tolonen. 2018. "African Mining, Gender, and Local Welfare." *European Economic Review* 101: 250–268.

Lei, Y.-H., and G. Michaels. 2014. "Do Giant Oilfield Discoveries Fuel Internal Armed Conflicts?" *Journal of Development Economics* 110: 139–157.

Lujala, P. 2010. "The Spoils of Nature: Armed Civil Conflict and Rebel Access to Natural Resources." *Journal of Peace Research* 47(1): 15–28.

Ross, M. L. 2001. "Does Oil Hinder Democracy?" *World Politics* 53(3): 325–361.

Ross, M. L. 2004. "What Do We Know About Natural Resources and Civil War?" *Journal of Peace Research* 41(3): 337–356.

Ross, M. L. 2012. *The Oil Curse: How Petroleum Wealth Shapes the Development of Nations.* Princeton, NJ: Princeton University Press.

Stock, J. H., and M. Yogo. 2002. "Testing for Weak Instruments in Linear IV Regression." Working Paper, National Bureau of Economic Research.

Sundberg, R., and E. Melander. 2013. "Introducing the UCDP Georeferenced Event Dataset." *Journal of Peace Research* 50(4): 523–532.

Vogt, M., N.-C. Bormann, S. Rüegger, L.-E. Cederman, P. Hunziker, and L. Girardin. 2015. "Integrating Data on Ethnicity, Geography, and Conflict." *Journal of Conflict Resolution* 59(7): 1327–1342.